COVID-19 Data Analysis

1. Data

Data Description:

The COVID database provides a time series of confirmed cases, deaths, recoveries, tests, vaccinations, hospitalizations, as well as the types of policy measures implemented to combat the pandemic. The data analysed in this project present statistics on the COVID-19 pandemic at the country level. Choosing a single country for analysis and prediction seems to be the only reasonable approach. As the dataset authors themselves note, aggregating data can lead to erroneous results, partly due to missing values. For this reason, data for different levels of detail are collected from different sources that provide information directly for the specified level.

Data Description:

Even when reviewing the documentation on the creation of the COVID dataset, it is possible to encounter numerous obstacles faced by the dataset creators. Due to the lack of international reporting standards and the use of various sources of information, international comparisons could not reliably provide meaningful conclusions. The creators, by constructing a detailed database, aimed to minimize potential inaccuracies in the collection of international data. It is worth noting that the data are standardized, but their originality is preserved, and no changes are made unless necessary for standardization—for example, calculating cumulative cases based on daily counts. Possible data inaccuracies are rarely corrected, but the dataset ensures verification is possible and provides access to the original data and code.

2. The Purpose and Value of Global and Local Analysis

2.1 Global analysis

After reviewing official reports on data collection methods and potential inaccuracies, one can conclude that a global analysis is unlikely to provide reliable insights. Several factors contribute to this.

First, comparing the number of infected individuals is largely dependent on the number of tests conducted. While the ratio of infected individuals to those tested could theoretically offer some indication, it also depends on which groups are being tested. For example, in extreme cases, if one country tests its entire population while another tests only people with typical COVID symptoms, such a ratio carries little meaningful information and can even be misleading.

Second, in some countries, COVID statistics were reported inconsistently—sometimes daily, sometimes weekly, etc. This inconsistency in reporting frequency can hinder trend analysis and lead to artificial spikes or unclear patterns. Missing data is also randomly distributed across different observations and features in many countries, which can similarly result in incorrect conclusions.

Finally, it is important to consider the political aspect of the pandemic. Media reports have highlighted potential data manipulation. Most discussions focus on underreporting in countries such as Russia and China, where such discrepancies may not have been accidental, but rather a result of broader political strategies.

2.2 Local analysis

Amid chaotic reports, missing data, and potential manipulations, it is still possible—after analysing the dataset and reviewing reporting methods and media reports—to identify countries that likely provided data relatively accurately reflecting the pandemic situation. Among these countries, Denmark appears to be one of the best choices for analysis. This selection was initially guided by several criteria outlined below.

1. Missing Data and Reporting Frequency

Table 1: Missing Values by Country, Sorted in Descending Order by 'percent_of_missing'

iso_alpha_3	percent_of_missing [%]	all_missing	rows_for_country
LTU	1.31	378	824
PER	4.02	952	677
CHL	5.01	1869	1066
PRT	9.03	2559	810
EST	9.11	3198	1003
SGP	11.46	2931	731
SVN	11.51	3615	897
DNK	12.79	5305	1185
JPN	12.98	5375	1183
BGR	13.04	4985	1092

Denmark (DNK) ranks 8th in the entire dataset in terms of the lowest percentage of missing values among all countries in the dataset (for this statistic, only features considered potentially useful were taken into account). Moreover, data for Denmark are recorded over a relatively long period of time, especially when compared to other European countries.

Table 2: Number of Missing Values for Selected Features, Among 1,185 Observations

	iso_alpha_3	date_missing	confirmed_missing	deaths_missing	recovered_missing	tests_missing	vaccines_missing
55	DNK	0	78	94	715	329	357

Apart from the considerable missing values in the "recovered" column—that is, for individuals who have recovered from the virus (which is also the case for many other countries)—the data appear to contain a lot of valuable information. First, most statistics in Denmark were reported daily, which allows for more detailed tracking of trends. Second, the dataset contains only a few randomly missing values. Looking at the distribution of NaNs for Denmark, many of the missing values are not random, mostly occurring at the beginning of the pandemic, when some statistics were naturally not yet available (such as the number of vaccinations).

2. Number of Conducted Tests

The number of tests conducted appears to be a key factor in accurately assessing infections. Therefore, an important reason for selecting Denmark for further analysis is the following data..

Tabela 3: Summary of Tests Conducted per Person, by Country, Sorted in Descending Order by „tests_per_person"

iso_alpha_3	tests	population	tests_per_person
CYP	29501771.0	1189265.0	24.806726
AUT	207903030.0	8840521.0	23.517056
ARE	168522672.0	9630959.0	17.498016
FRO	777845.0	48497.0	16.039033
GIB	533268.0	33718.0	15.815529
DNK	64649913.0	5793636.0	11.158781
SVK	51238482.0	5446771.0	9.407130
SVN	17197882.0	2073894.0	8.292556
GRC	84483362.0	10731726.0	7.872300
GBR	514605757.0	66460344.0	7.743050

It can be seen here that Denmark again ranks among the top 10, this time in terms of the number of tests conducted per capita.

3.Offical analysis

The table above showed that more than 11 tests were conducted per person in Denmark. Analysis of data from the National Library of Medicine (a U.S. government website considered a reliable source) in the article *"Patterns of testing in the extensive Danish national SARS-CoV-2 test set-up"* summarizes that "during the studied period (27 February 2020 to 10 March 2022), a total of 63.7 million PCR tests and 60.0 million rapid antigen tests were conducted, testing at least once 90.9% and 78.8% of the Danish population, respectively, through PCR and antigen tests." The number of tests reported on this website aligns with the information in the dataset.

Combining these sources, we can conclude that most of the population was tested at least once, although some individuals were tested multiple times (for example, healthcare workers). The text also notes that "the Danish national SARS-CoV-2 mass testing system was one of the most ambitious in the world." The aforementioned report, along with several other sources, generally emphasizes the strong methodology of COVID data collection in Denmark, which was also an important criterion in choosing this country for analysis.

In the next subsection, data from around the world will be analyzed, with particular focus on Europe and Denmark. This will help confirm that Denmark is suitable for further analysis and will provide more detailed comparisons regarding the course of the pandemic in the selected country.

3. Exploratory Data Analysis with a Focus on Europe and Denmark

Initially, the analysis focuses on the features defined in the documentation as epidemiological variables, namely confirmed cases of infection, deaths, recoveries, number of tests conducted, number of vaccinated individuals, and the number of people vaccinated with each dose.

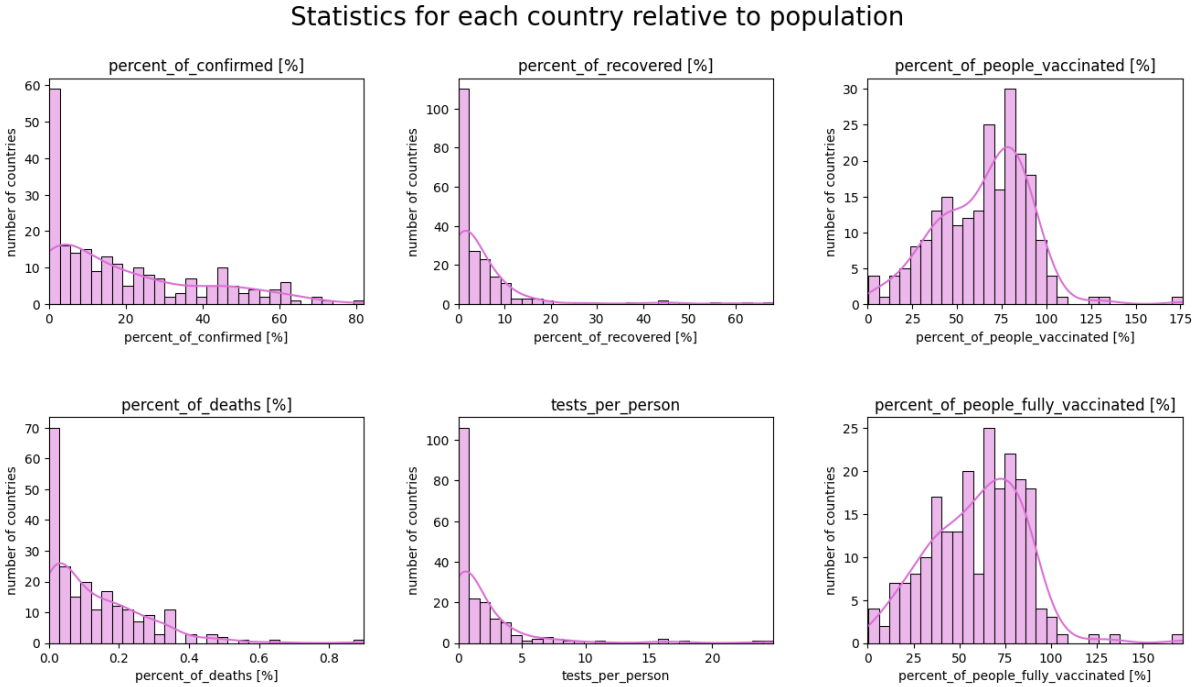

Figure 1: Selected Epidemiological Variables Grouped by Country

From the charts above, it can be observed that most countries showed a relatively small percentage of citizens infected with the virus. However, this distribution closely mirrors the distribution of tests conducted, suggesting that many countries likely underreported the number of cases.

For variables related to the number of vaccinated individuals, something less obvious—and even incorrect—can be noticed: the percentage of vaccinated people exceeds 100%, suggesting that more than 100% of the population has been vaccinated. This statistic was calculated by comparing the maximum cumulative number of vaccinated individuals to the total population, so such a value most likely results from erroneous data reporting. This type of problem can be observed in 8 countries:

Table 4: Percentage of Vaccinated Individuals by Country, Sorted in Descending Order

	iso_alpha_3	people_vaccinated	population	percent_of_people_vaccinated [%]
0	COK	1.511200e+04	8.574000e+03	176.253791
1	TKL	2.203000e+03	1.647000e+03	133.758349
2	GIB	4.217500e+04	3.371800e+04	125.081559
3	MAC	6.797030e+05	6.316360e+05	107.609921
4	BRN	4.511490e+05	4.289620e+05	105.172253
5	ARE	9.991089e+06	9.630959e+06	103.739295
6	NRU	1.310600e+04	1.270400e+04	103.164358
7	QAT	2.852178e+06	2.781677e+06	102.534478

For further international statistics, these countries are therefore excluded due to the obvious errors in the collected data.

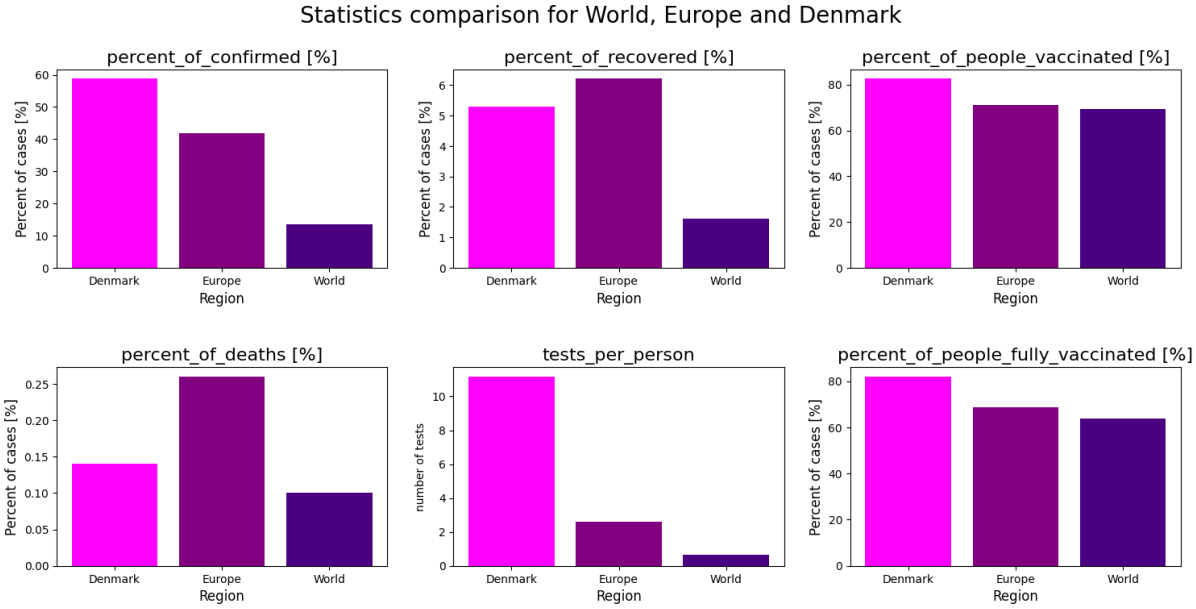

Figure 2: Comparison of Statistics for the World, Europe, and Denmark, Accounting for Population

From the charts above, it can be seen that Denmark significantly exceeds both Europe and the world in terms of the number of tests conducted and also surpasses European countries and the global average in the percentage of the population that has been partially or fully vaccinated. Although the percentage of confirmed cases is much higher than in Europe or worldwide, the percentage of the population that has died from COVID-related complications is lower than in Europe. For the world, this percentage is lower as well, although missing values—present in a significant portion of national reports across many observations—may influence this result.

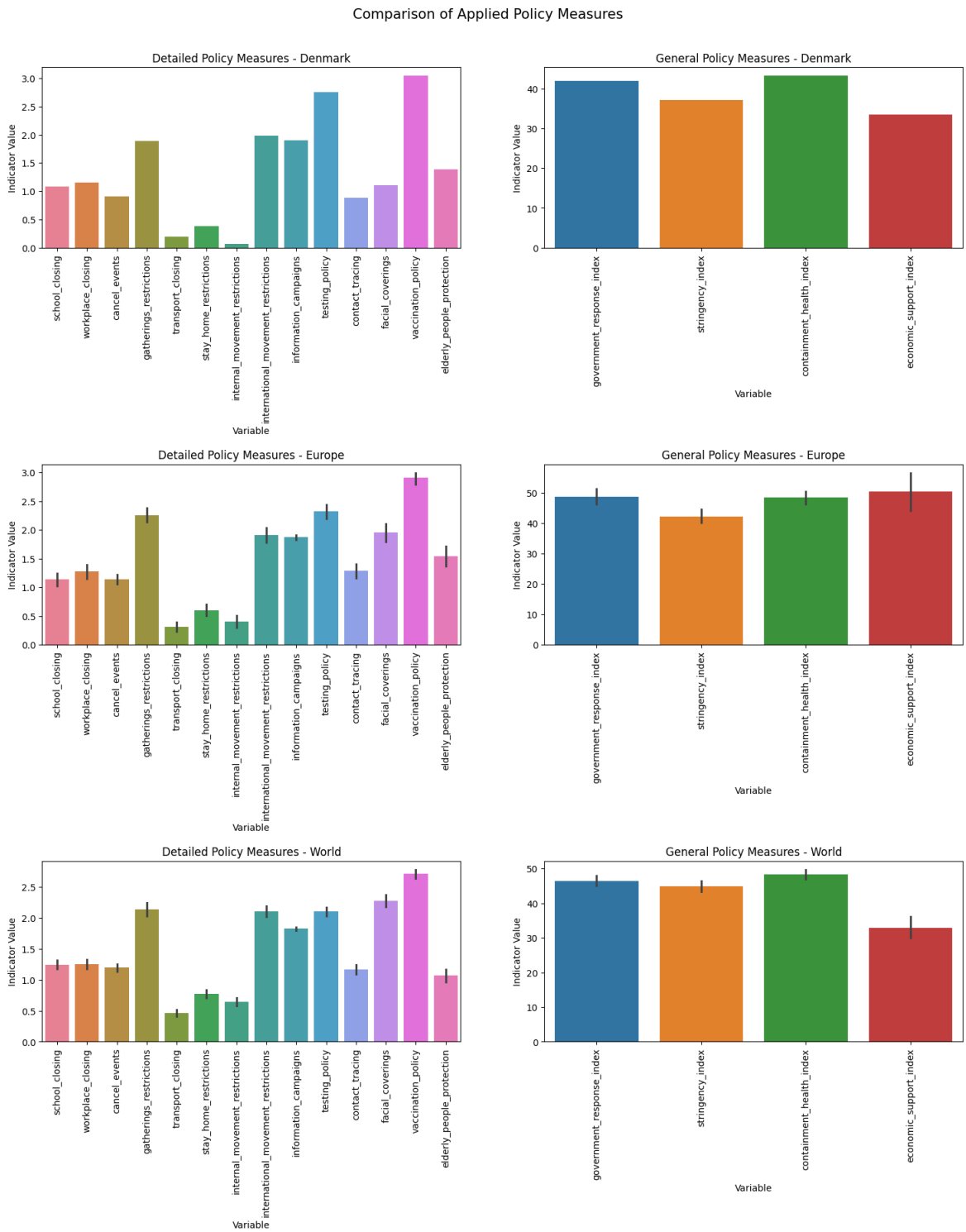

Figure 3: Comparison of Policy Measure Indicators

The charts comparing pandemic-related policy measures can, with some generalization, be interpreted as follows: the higher the value, the more stringent the measures implemented for a given issue. In the adjacent chart, we can observe that most countries placed varying degrees of emphasis on similar areas. What again distinguishes Denmark positively is its higher indicator for testing-related policies compared to broader regions. This further supports earlier observations regarding the potential reliability of results in terms of the actual number of infected individuals in the country.

4. Variables and missing values

4.1. Variable selection

Since the further analysis will focus on a specific country—Denmark—variables related to administrative divisions and similar categories are no longer needed. Additionally, all variables related to external keys can also be removed during the model creation stage. Furthermore, the policy measures applied to different public spaces or social groups can be limited to more general descriptive statistics. Ultimately, they were reduced to only two indices based on the indicators used to calculate them. In the model, two variables are used whose components do not overlap:

- **Containment and Health Index -** considers restriction and lockdown policies as well as health care-related policies
- **Economic Support Index -** considers only economic measures

4.2 Missing values

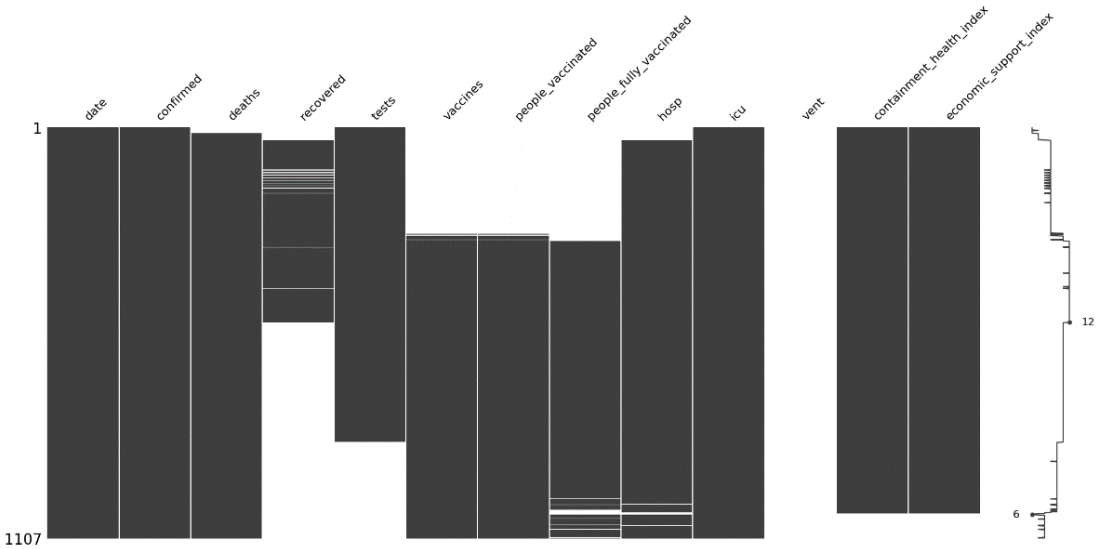

Figure 4: Distribution of Missing Values Generated Using the missingno Library

Initially, the dataset can be limited to observations for which the number of confirmed cases ("confirmed") is not missing. This corresponds to the period from the first recorded case to the point where such cases were no longer reported or no longer exist in the dataset.

After this operation, the number of deaths is missing for the initial phase of the pandemic. It can therefore be assumed that there were no fatalities at that time, and NaN values can be replaced with 0.

A similar assumption can be applied to columns related to vaccinations and hospitalizations ("hosp" and "icu"). All cells before the first recorded vaccinations or hospitalizations will be set to zero.

The variable indicating the number of people who have recovered from the virus contains many missing values, suggesting that tracking this statistic was not a priority during periods of high infection. Furthermore, it can be assumed to some extent that if someone did not die from the infection, they must have recovered. With this assumption, this column is not needed at all and was therefore completely removed from the dataset used in the model.

The "vent" variable, for the dataset limited to Denmark, is empty and is therefore entirely removed.

After these preliminary changes, the distribution of missing values looks as follows:

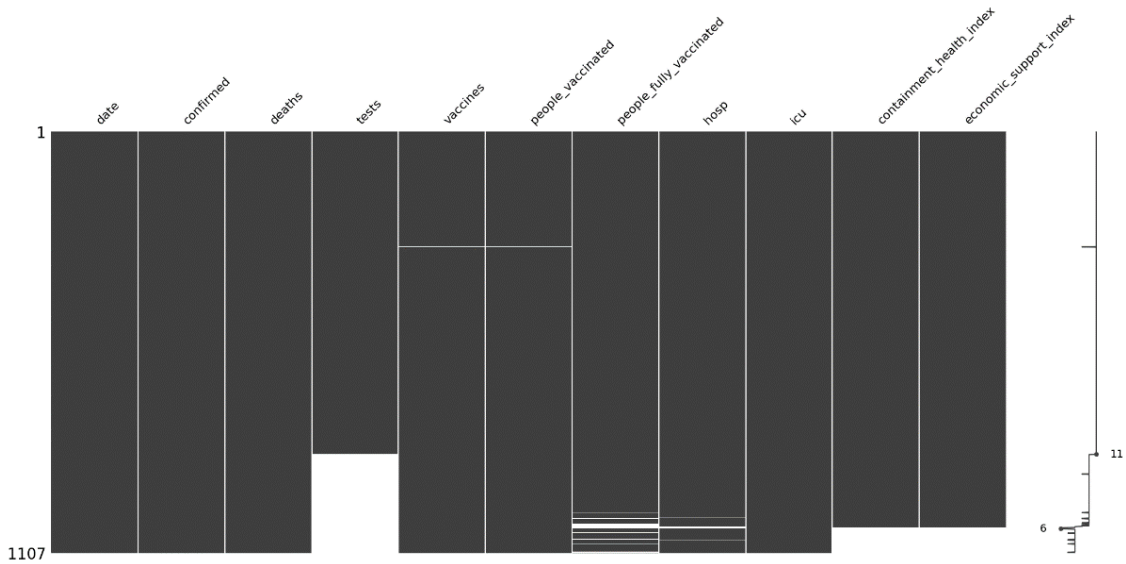

Figure 5: Distribution of Missing Values Generated Using the missingno Library

Several methods were tested after building the models, including using the mean, median, interpolation, forward fill, and linear regression.

For all columns except the one related to tests, missing values were filled using interpolation, which is theoretically well-suited for time series data and, in fact, provided the best results among all tested methods.

For the "tests" variable, the metrics for different methods were generally similar, although slightly better results were obtained using forward fill or linear regression. Ultimately, the missing values for this variable were filled using the last observed value (forward fill).

5. Linear Regression Model

In the linear regression model, the previously selected variables were used, and the date was recorded as the number of days elapsed since the first detected case.

When analysing the metrics for specific variables, MSE, RMSE, MAE, and R^2 were used. Primarily, the selection of the best variable is based on RMSE (which indicates how well the regression model can predict the variable's value) and R^2 (which indicates how well the predictor variables explain the variability).

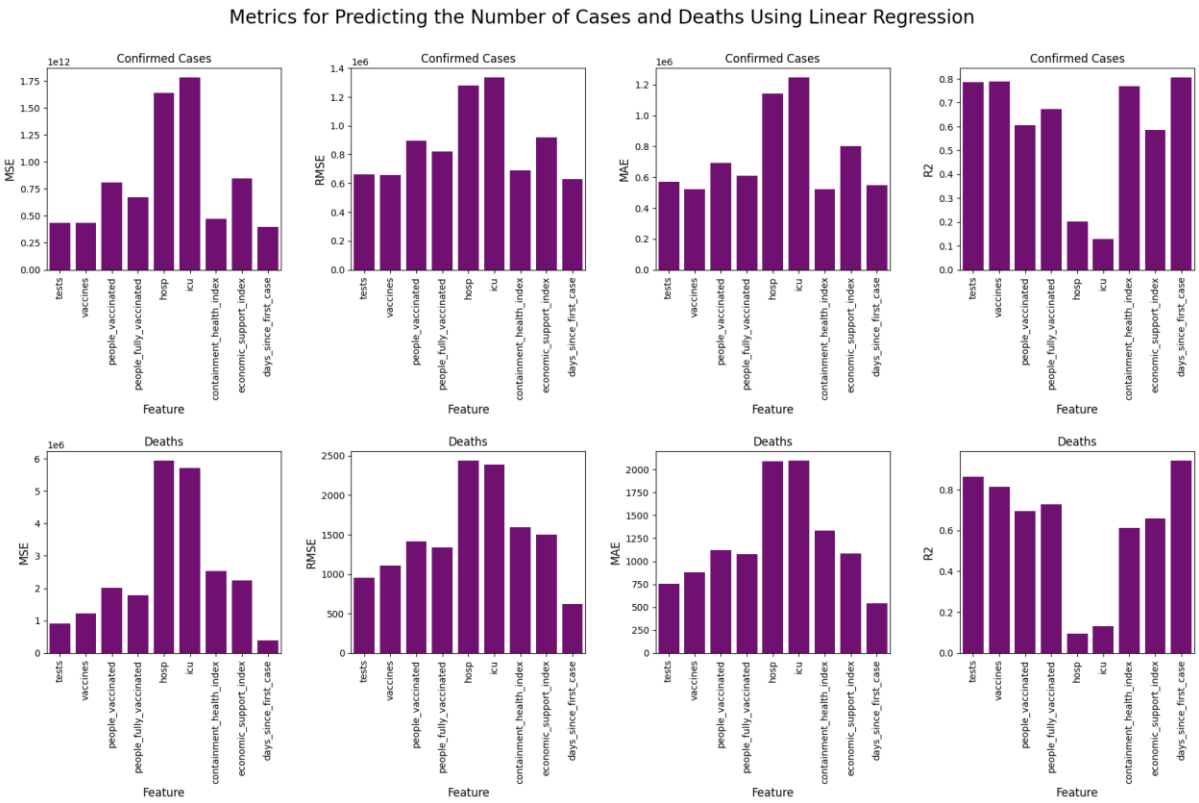

Figure 6: Metric Plots for Explanatory Variables in the Linear Regression Model Predicting Cases and Deaths

6. Linear Regression Assumptions

For predicting confirmed cases, three variables stand out as having the best values for the considered metrics (relatively low RMSE and high R^2): „tests", „vaccines" and „days_since_first_case". To check the assumptions of linear regression, the variable „vaccines" - representing the total number of vaccine doses administered—is used.

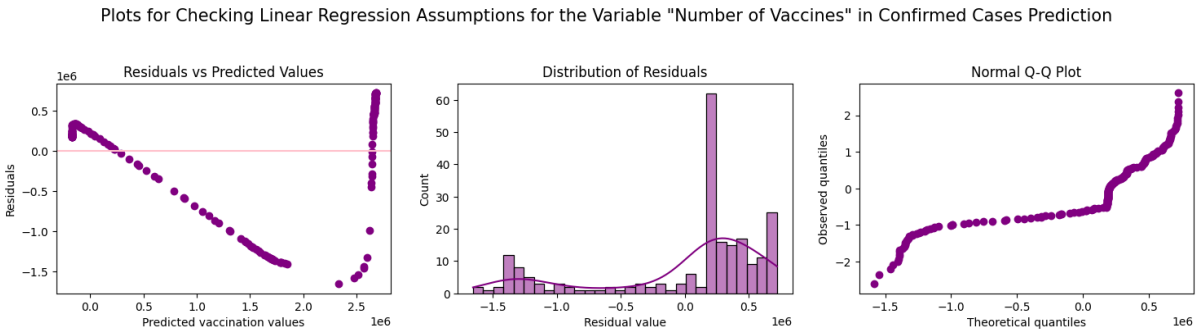

Figure 7: Plots for Checking Linear Regression Assumptions for the "Vaccines" Variable in Predicting Confirmed Cases

Based on the charts above, we can conclude that even the theoretically best variable does not meet the assumptions of linear regression. In the first plot, it can be seen that the residuals are not evenly distributed around the line, which violates the homoscedasticity assumption and suggests a problem with the linearity of the model.
In the second plot, the residual distribution does not resemble a normal distribution at all, showing two local peaks and narrowing toward lower values in the middle of the distribution, meaning another homoscedasticity assumption is not satisfied.
In the last plot, the pattern is flattened in the middle and deviates significantly at the extremes, further confirming the lack of normality in the residuals.

For predicting the number of deaths, one variable stands out with the best metric values, showing the highest R^2 and the lowest values for the other metrics. This variable is the number of days elapsed since the first detected case.

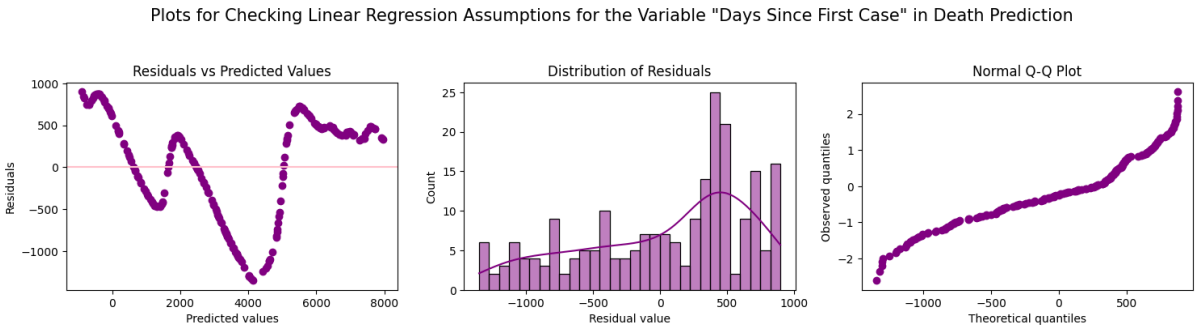

Figure 8: Plots for Checking Linear Regression Assumptions for the "Days Since First Case" Variable in Predicting Deaths

Similarly to the prediction of confirmed cases, the best variable for predicting deaths does not satisfy the assumptions of linear regression. In the first plot, the residual distribution appears patterned and uneven around the y = 0 axis. The second plot resembles a normal distribution slightly more than in the previous model, but the distribution is still asymmetric, meaning the residuals are not normally distributed, which is also suggested by the last plot.

7. Multicollinearity of Variables and Multiple Regression

Multicollinearity of variables is mainly checked using the correlation matrix and by considering previous results.

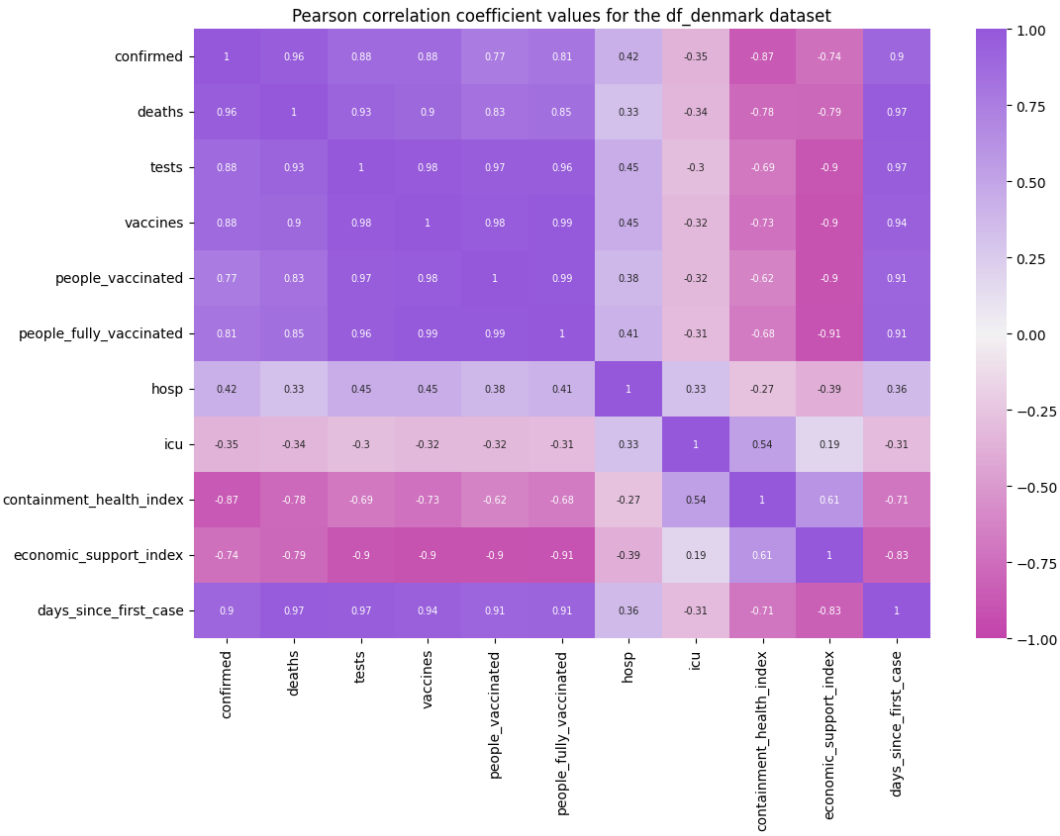

Figure 9: Macierz wartości współczynnika korelacji Pearsona dla danych odnośnie Danii

Among all the variables used, the variables „hosp” and „icu”- representing the number of hospitalized patients and the number of patients in intensive care, respectively - stand out with significantly lower correlation coefficients compared to the others. These variables will be temporarily removed to test and compare the prediction accuracy.

Table 5: Metrics including „hosp” i „icu”

	Predicted_variable	MSE	RMSE	MAE	R2
0	CONFIRMED	4.744371e+10	217815.765414	164836.606161	0.976813
1	DEATHS	1.189350e+05	344.869548	267.786738	0.981853

Tabela 6: Metrics without including „hosp” i „icu”

	Predicted_variable	MSE	RMSE	MAE	R2
0	CONFIRMED	4.744371e+10	217815.761632	164836.586843	0.976813
1	DEATHS	1.600224e+05	400.027944	307.535340	0.975584

From the tables above, it can be seen that including these two variables improves the performance of both models, although the effect is much more significant for predicting deaths. Therefore, the decision was made to include the **“hosp”** and **“icu”** variables in the predictive models for deaths, while omitting them for predictions of confirmed cases. This conclusion is also quite logical when considering the meaning of the variables: if someone was in sufficiently poor condition to require hospitalization, the likelihood of death increases significantly.

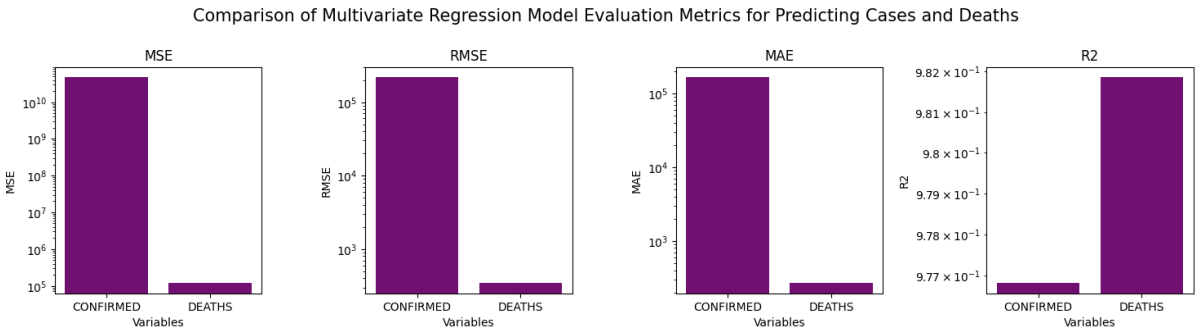

Figure 10: Evaluation Metrics for the Multiple Regression Model Predicting Cases and Deaths (Using a Logarithmic Scale)

In the chart above, it is seen that the multiple regression model (as with the single-variable linear regression model) performs better at explaining variability and is significantly more accurate in predicting values for the „deaths” variable than for „confirmed” cases.

Table 7: Comparison of Metric Values for Multiple Regression and for the Best Single-Variable Linear Regression

	Regression type	Predicted_variable	MSE	RMSE	MAE	R2
0	linear regression for vaccines	CONFIRMED	4.339991e+11	658786.046853	522076.937050	0.787890
1	multiple regression	CONFIRMED	4.744371e+10	217815.765414	164836.606161	0.976813
2	linear regression for days_since_first_case	DEATHS	3.909911e+05	625.292839	540.168417	0.940344
3	multiplle regression	DEATHS	1.189350e+05	344.869548	267.786738	0.981853

In the table above, we can observe an improvement in predictions for the multiple regression model for both variables, with the improvement being significantly more pronounced for predicting the number of confirmed cases.

8. Comparison of Different Regression Algorithms

Table 8: Metrics for Predicting the confirmed" and „deaths" Variables Using Different Algorithms

	Regression_model	Predicted_variable	MSE	RMSE	MAE	R2
0	LinearRegression for vaccines	CONFIRMED	4.339991e+11	6.587860e+05	5.220769e+05	0.787890
1	LinearRegression for days_since_first_case	DEATHS	3.909911e+05	6.252928e+02	5.401684e+02	0.940344
2	MultipleRegression	CONFIRMED	4.744371e+10	2.178158e+05	1.648366e+05	0.976813
3	MultipleRegression	DEATHS	1.189350e+05	3.448695e+02	2.677867e+02	0.981853
4	SVR	CONFIRMED	2.884829e+12	1.698478e+06	1.144947e+06	-0.409911
5	SVR	DEATHS	2.884829e+12	1.698478e+06	1.144947e+06	-0.409911
6	DecisionTree	CONFIRMED	8.337361e+07	9.130915e+03	3.261842e+03	0.999959
7	DecisionTree	DEATHS	8.535004e+07	9.238509e+03	3.293635e+03	0.999958
8	RandomForest	CONFIRMED	1.883142e+07	4.339518e+03	1.281341e+03	0.999991
9	RandomForest	DEATHS	1.413716e+07	3.759942e+03	1.160476e+03	0.999993

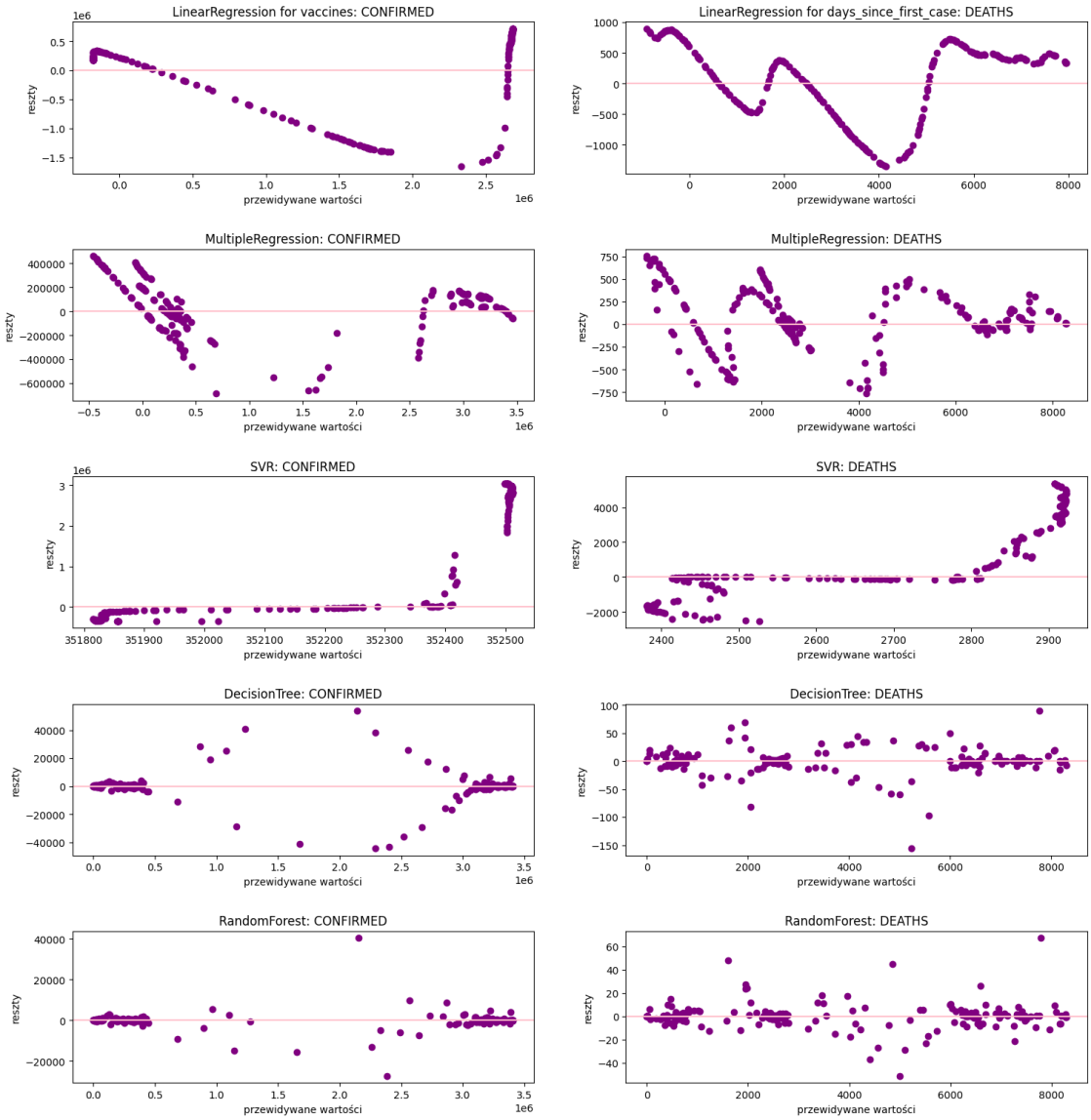

Figure 11: Distribution of Residual Values for the Tested Models

To compare the predictive accuracy of the tested models, previously used metrics and residual distribution plots relative to predicted values were employed.

Simple and Multiple Linear Regression Models: The issue of poor fit for these models was already discussed in an earlier subsection, but it is worth highlighting the analysis of metrics and residuals here. A high R^2 value for the multiple regression model might suggest accurate predictions; however, the residual distribution clearly contradicts this conclusion. The residuals are not randomly distributed—they follow a pattern in certain areas, alternately overestimating and underestimating the true values. This may indicate that the model lacks linearity, suggesting that more complex regression models are needed.

SVR Algorithm: For the model built using the SVR (Support Vector Regression) algorithm, the R^2 metric alone indicates that the model does not follow the trend of the predicted variable and is clearly a poor choice. Properly tuned parameters might improve the performance of this type of regression model, but leaving them at default is a very weak choice for this kind of data.

Regression Tree Algorithm: In this type of algorithm, noticeable clusters of residuals appear around extreme predicted values, particularly for predicting the number of cases, which may indicate that the model does not generalize well for intermediate values. Again, a high R^2 score can be misleading, potentially suggesting poor generalization and overfitting.

Random Forest Algorithm: Among all the analysed algorithms, this one seems to perform best in predicting the variables. This is suggested by an R^2 value close to one, the lowest RMSE and other metric values, and supported by the residual distribution plots. For both predicted variables, clusters of residuals near zero are still visible; however, in this case, no discernible "pattern" can be identified—the distribution appears completely random. Overall, this algorithm seems to be the most accurate among those analysed, and even using only default parameters, it is the best choice for this dataset.

9. Conclusions

Regression-Based Predictive Algorithms for COVID Data

After testing various regression models, it can be observed that the quality of predictions largely depends on selecting the appropriate algorithm for the problem being analysed. Another important factor in choosing the optimal model is the proper tuning of parameters, which can further improve already good predictions. Overall, with well-prepared and properly selected data, predicting cases and deaths can be achieved with fairly good accuracy using regression algorithms. On the other hand, there are several factors that highlight potential challenges with this approach. The most important seems to be the issue of autocorrelation in data such as these, i.e., time series data where past values influence subsequent ones and observations are not independent. Many sources suggest alternative models for time series analysis, such as ARIMA.

Global Analysis and Prediction of Data

While performing predictions on a local scale makes it difficult to conclusively assess whether a global approach is optimal, it seems plausible. However, as discussed earlier in this work, large gaps in data, differences in reporting across countries, and potential data manipulation pose challenges for creating a model that can accurately reflect reality. Setting these issues aside, a global model could still provide important insights, for example, in tracking the virus's spread patterns or comparing well-designed policy measures to effectively mitigate pandemic effects. In summary, global analysis and prediction could yield valuable observations and conclusions, but more measurable and reliable insights are likely to come from more specific regions, especially those that reported pandemic data most accurately.

Assessment of Deaths and Cases

General observations regarding deaths and cases were already discussed in the EDA section. It is worth emphasizing the clear seasonality characterizing these variables, as well as significant differences in the proportion of the population infected with the virus versus those who died as a result. Across different countries, the COVID situation likely evolved very differently, which is also linked to many other variables. However, it should be noted that the raw statistics do not necessarily perfectly reflect the real situation, especially globally during the COVID pandemic.

Sources:

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